

The empirical recurrence for OEIS A250711, recovered from its tail

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Abstract

OEIS A250711 records “Empirical recurrence of order 93” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 490$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 101$. Its last coefficient is $c_{93} = 10077696000 \neq 0$, so no further tail satisfies anything shorter; any order-93 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A250711 is “Number of $(4+1)X(n+1)$ 0..3 arrays with nondecreasing $x(i,j)+x(i,j-1)$ in the i direction and nondecreasing $\min(x(i,j),x(i-1,j))$ in the j direction.”. It has offset 1 and begins

10050, 160486, 2003383, 19962956, 171212264, 1307242006, 9230232528, 61126170276, ...

The entry states:

Empirical recurrence of order 93 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:44:06, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 490$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 490$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 93: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 93.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 980$ exact terms from $n = 8$ on, it returns order 93 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	42	c_{25}	32844532937469097	c_{49}	2495722432587315542587	c_{73}	−11465518
c_2	−721	c_{26}	179048800495795887	c_{50}	22282038999912742611977	c_{74}	10461517
c_3	5716	c_{27}	−410057348237295451	c_{51}	−14729892216067972877550	c_{75}	5498632
c_4	−5274	c_{28}	−411026247747342559	c_{52}	−20330081556732131701194	c_{76}	−2910267
c_5	−293228	c_{29}	2443727273028233042	c_{53}	27096493279496122910216	c_{77}	5890516
c_6	2315495	c_{30}	−893813797103866283	c_{54}	9834352742942129528667	c_{78}	5087126
c_7	−2408829	c_{31}	−9329033871571001045	c_{55}	−32820616388788563621954	c_{79}	−2405692
c_8	−64155245	c_{32}	13105384950985704484	c_{56}	4762541171308494176865	c_{80}	−396246
c_9	364567010	c_{33}	21788879758119835195	c_{57}	28554421677987834827293	c_{81}	4957309
c_{10}	88633101	c_{34}	−66376092684324338258	c_{58}	−16063389853276108002438	c_{82}	−475258
c_{11}	−8288830895	c_{35}	−11620642682409965204	c_{59}	−17185784322633014631571	c_{83}	−578119
c_{12}	26300523595	c_{36}	215918969627467083160	c_{60}	19299350922627220976731	c_{84}	1773476
c_{13}	62384539745	c_{37}	−146654226857729527255	c_{61}	5284800965633234148269	c_{85}	266476
c_{14}	−588190966060	c_{38}	−476938142021213386442	c_{62}	−15332342566026658155380	c_{86}	−21903
c_{15}	667219224451	c_{39}	750832959524595387067	c_{63}	2087224840181776375068	c_{87}	199214
c_{16}	5924694330499	c_{40}	609391823865301255636	c_{64}	8660761243425033243207	c_{88}	114436
c_{17}	−21325864372813	c_{41}	−2177524751259542433838	c_{65}	−4157921200180522234134	c_{89}	−3188
c_{18}	−18226677804291	c_{42}	179343552553798059108	c_{66}	−3294586361803828202706	c_{90}	2153
c_{19}	247059929298881	c_{43}	4377952754843003427311	c_{67}	3169667494765804421482	c_{91}	11098
c_{20}	−298155499439555	c_{44}	−3183651239072203893381	c_{68}	563517719487119358550	c_{92}	−184
c_{21}	−1550972607323877	c_{45}	−6082267601387860046912	c_{69}	−1577457926853868091275	c_{93}	1007
c_{22}	4916480154810414	c_{46}	9225302652885676665550	c_{70}	237524162604459168249		
c_{23}	3442846205491835	c_{47}	4623434462454697890803	c_{71}	539890903807895099156		
c_{24}	−38233251172953457	c_{48}	−16976581949469654911726	c_{72}	−238213676762599218518		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 101$, and it is the recurrence OEIS A250711 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{93} - c_1x^{93-1} - \dots - c_{93}$. Its constant term is $-c_{93} = -10077696000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 93 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 93, so $\deg q = 93$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 93, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 93, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 10077696000.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-93 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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